Motorcycle Helmet Detection Using YOLOv8

Final Project Report

This project develops an object-detection model to identify motorcycle riders wearing helmets and riders without helmets.

1. Project Objectives

The objective is to train and evaluate a YOLOv8 object-detection model using two classes: With Helmet and Without Helmet.

2. Dataset Description

Dataset split	Images
Training	140
Validation	40
Testing	20
Total	200
Classes	With Helmet; Without Helmet
Valid annotated boxes	857

The dataset was organized into training, validation and testing splits using a 70/20/10 distribution. Dataset structure and label format were programmatically checked.

3. Model and Training Configuration

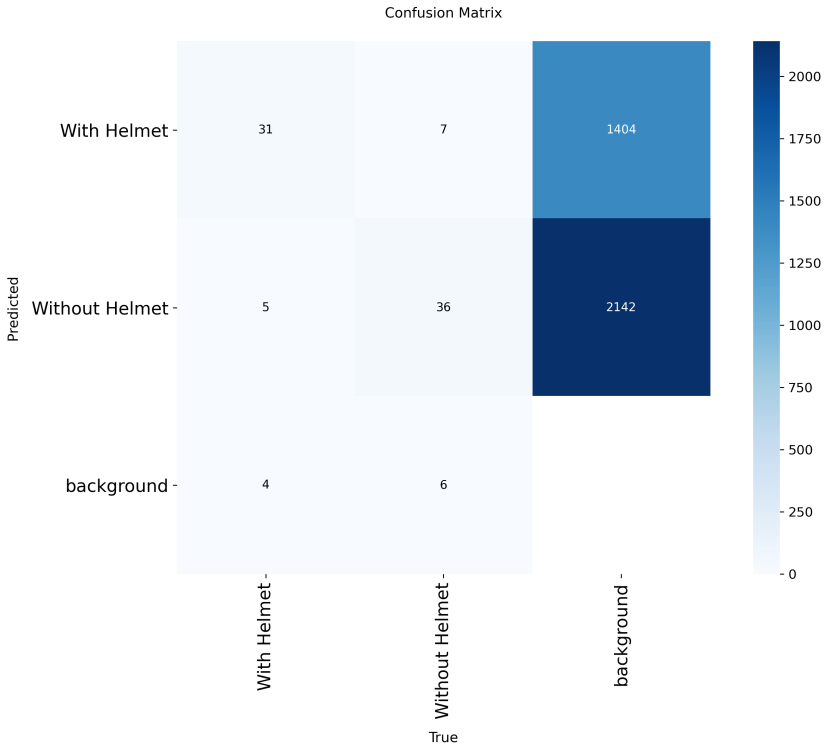
Model: YOLOv8n (pretrained weights)
Image size: 640 × 640
Epochs: 30
Batch size: 8
GPU: NVIDIA Tesla T4
Training time: approximately 4.27 minutes
Ultralytics version: 8.4.211

4. Test Set Evaluation

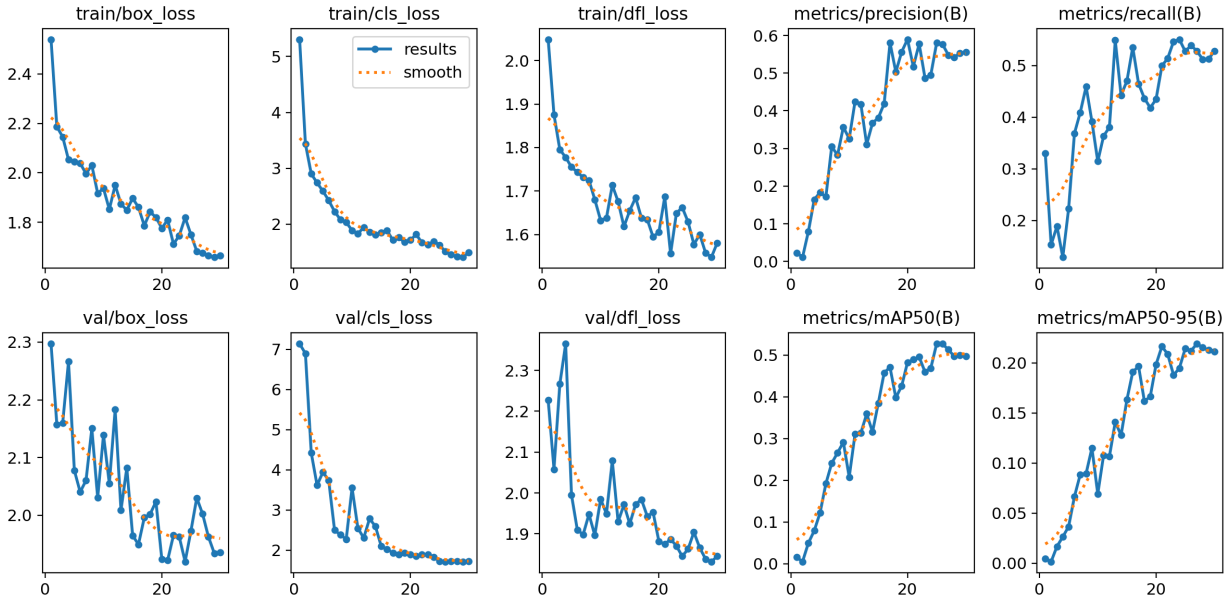
Metric	Test result
Precision	69.8%
Recall	59.6%
mAP@0.5	65.0%
mAP@0.5:0.95	27.7%

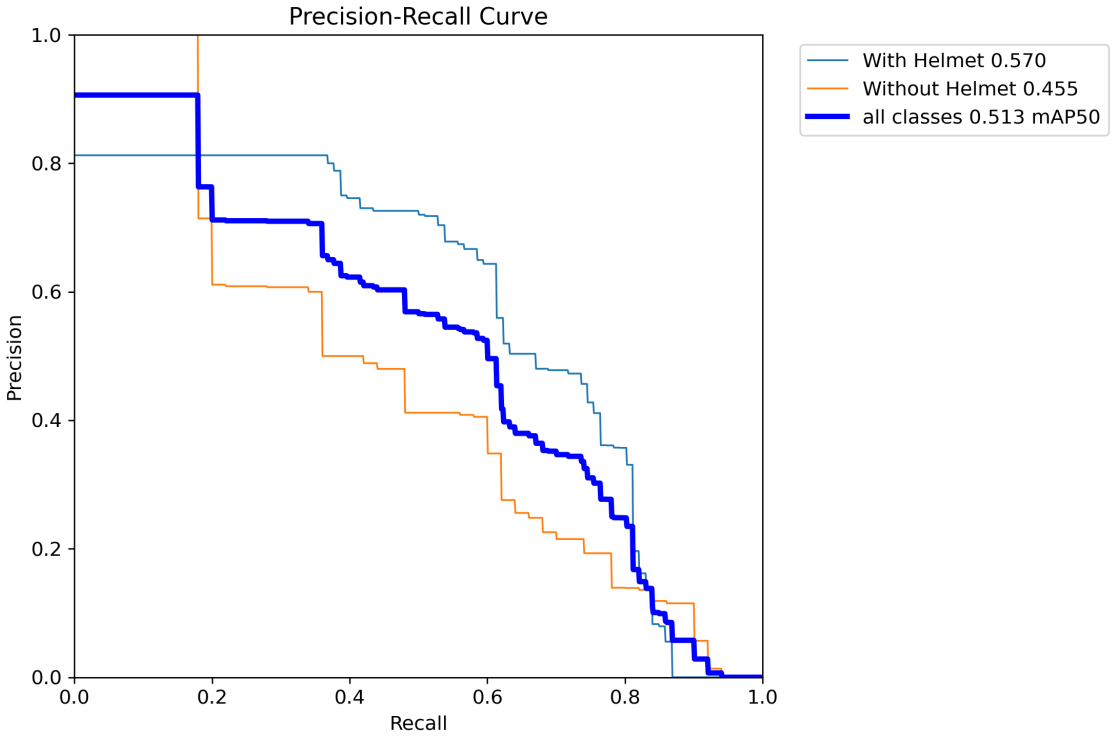
Class-wise test results:
With Helmet: Precision 57.1%, Recall 60.0%, mAP@0.5 62.9%.
Without Helmet: Precision 82.5%, Recall 59.2%, mAP@0.5 67.2%.

5. Confusion Matrix



6. Training Curves





7. Prediction Results

The trained model was applied to all 20 images in the test split. Annotated prediction images containing predicted classes, bounding boxes and confidence scores were saved in outputs/predictions.

manual__BikesHelmets115_png_jpg.rf.e2d44812a061abe2376fea5b3f9dcdc6.jpg



manual__BikesHelmets118_png_jpg.rf.276fb4c4dc04d2bb205f3790db6968ff.jpg



manual__BikesHelmets128_png_jpg.rf.689974532482e03a0a3b4362a02c8d3b.jpg



manual__BikesHelmets147_png_jpg.rf.325efbcc48fe92642606c8249b03f457.jpg



8. Automated Error Analysis

Error category	Count
Correct detection	32
False Negative	13
False Positive	12
Wrong class	7
Total records	64

These counts were produced by an automated comparison of predictions with ground-truth labels on the first 10 test images, using an IoU matching threshold of 0.5. They are detection records, not image counts. A separate manual visual review of 10 incorrect or weak predictions has not yet been completed.

9. Challenges and Limitations

The test results show that the model does not detect every annotated object. False positives, missed detections and wrong-class predictions were observed in the automated comparison. The dataset is relatively small, and the model's performance may be affected by image quality, object size, lighting, occlusion and annotation quality. These are possible factors and require further visual investigation.

10. Proposed Improvements

1. Add more diverse and accurately annotated training images.
2. Review bounding boxes and class labels to improve annotation quality.
3. Compare larger YOLO models and tune training settings, including epochs, batch size and confidence threshold.

11. Project Deliverables

The project folder contains the dataset, training notebook, trained best.pt model, training outputs, test evaluation outputs, 20 prediction images, automated error-analysis CSV, README.md and requirements.txt.

12. Conclusion

A YOLOv8n helmet-detection model was trained for 30 epochs and evaluated on a separate test split. The test mAP@0.5 was 65.0%. The project demonstrates an end-to-end object-detection workflow, while also identifying areas that need further improvement.

Correction and Update: Manual Error Review

A visual review was performed on the error-heavy test image. Ten error records from this image were examined alongside the actual annotations and model predictions.

Error Type	Reviewed Records
Wrong Class	2
False Positive	4
False Negative	4
Total	10

Possible explanations include class confusion, unmatched detections, and missed objects. These are potential explanations; the exact causes were not independently established.

Proposed Improvements

• Increase and balance the training dataset.

• Improve bounding-box and class-label annotation quality.

• Tune training and inference settings using validation data.

Error Analysis and Manual Review

Automated error analysis identified 32 incorrect detection records among the evaluated samples: 13 False Negatives, 12 False Positives, and 7 Wrong Class predictions.

Manual Review: 10 Incorrect Predictions

Image	Actual	Prediction	Error Type	Possible Reason
BikesHelmets115_png_jpg.rf.e2d44812a061abe2376fea5b3f9dcdc6	Without Helmet	With Helmet	Wrong class	Model assigned an incorrect helmet-status class.
BikesHelmets115_png_jpg.rf.e2d44812a061abe2376fea5b3f9dcdc6	Without Helmet	With Helmet	Wrong class	Model assigned an incorrect helmet-status class.
BikesHelmets115_png_jpg.rf.e2d44812a061abe2376fea5b3f9dcdc6	No matched object	With Helmet	False Positive	Model detected an object where no matching ground-truth object exists.
BikesHelmets115_png_jpg.rf.e2d44812a061abe2376fea5b3f9dcdc6	No matched object	With Helmet	False Positive	Model detected an object where no matching ground-truth object exists.
BikesHelmets115_png_jpg.rf.e2d44812a061abe2376fea5b3f9dcdc6	No matched object	With Helmet	False Positive	Model detected an object where no matching ground-truth object exists.
BikesHelmets115_png_jpg.rf.e2d44812a061abe2376fea5b3f9dcdc6	No matched object	With Helmet	False Positive	Model detected an object where no matching ground-truth object exists.
BikesHelmets115_png_jpg.rf.e2d44812a061abe2376fea5b3f9dcdc6	Without Helmet	Not detected	False Negative	Model failed to detect an annotated object.
BikesHelmets115_png_jpg.rf.e2d44812a061abe2376fea5b3f9dcdc6	Without Helmet	Not detected	False Negative	Model failed to detect an annotated object.
BikesHelmets115_png_jpg.rf.e2d44812a061abe2376fea5b3f9dcdc6	Without Helmet	Not detected	False Negative	Model failed to detect an annotated object.
BikesHelmets115_png_jpg.rf.e2d44812a061abe2376fea5b3f9dcdc6	Without Helmet	Not detected	False Negative	Model failed to detect an annotated object.

Recommended Model Improvements

1. More Diverse Training Data

Add more annotated images with different camera angles, lighting conditions, crowded scenes, and motorcycle backgrounds.

2. Annotation Quality Improvement

Carefully review person and helmet bounding boxes and correct inaccurate or inconsistent annotations.

3. Data Augmentation and Model Tuning

Use suitable augmentation for blurry, small-object, and low-light images. Experiment with confidence thresholds and inference settings to reduce false detections.